

Track 1 evidence review: authorization, observation and preview fidelity

13 September 2026

Authoring material, not a submission manuscript. This is the consolidated, AI-assisted evidence package through H33. The author must review the facts and write the final narrative in the [official template](#). The original `evidence-pack.pdf` and `review.html` are historical renderings; `track1-review.pdf` and `track1-review.html` render this current source. Nothing in this package has been submitted or independently reproduced by a reviewer.

Author review needed: Aditya Pratap Singh; affiliation unconfirmed. Working report title: *Where authorization stops: auditable containment and its limits*. The [event guidelines](#) and template govern the final submission. Use at most 150 words for the abstract, at most eight main pages, and include the limitations/dual-use appendix.

Findings to review

1. **Execution-time authorization helps in the tested environments.** In native Git, an equally informed broker preview allowed 3/6 forbidden ref-update workflows; the receive-time effect gate allowed 0/6. Both completed 5/5 benign workflows.
2. **Independent backend evidence reveals effects upstream records miss.** On six executed Pome attacks, intent detected 0/6, gate-entry evidence 2/6 and backend tape 6/6. Detection did not undo those effects.
3. **Quarantine depends on preview fidelity.** Selective release blocked four modeled fault classes while preserving benign work, but every simulation-gap repetition escaped. The execution-only counterexample also persists in the later hosted and separate-process local studies.
4. **The execution boundary is where enforcement belongs.** An execution-time effect gate on the service path prevents the execution-only write that static and preview miss (§4.5), and broadening it to confine outbound data flow also stops a read exfiltration all three miss (§4.6) — reinforcing finding 3, not overturning it.

These are separate experiments and denominators. They do not establish production security, prevention of the actual incident, or quarantine superiority against a realistic AI attacker. H27 is incomplete, H30 lacked substantive attacks, and H31 failed its search-coverage gate. Their unfavorable outcomes remain in Section 4.4.

1. Introduction

Argument for the author to develop: containment needs evidence of both the authority checked and the effect actually executed. An accurate intent log can miss changes later in the call path. A correct dry run can also disagree with a later execution. The deliverable is an engineering control/evidence checklist, backed by reproducible examples of prevention, detection and failure.

The intended reader is an agent-platform or evaluation engineer deciding where to place enforcement and what evidence to retain. Track 1 fit rests on inspectable requirements and practical review, not on obtaining a positive quarantine result. No lab has assessed adoption, and no external reviewer has reproduced this package.

Preserved research question

Can an adaptive effect-bound broker reduce unauthorized agent effects and detect execution mismatches better than intent logging or static authorization, while keeping legitimate actions usable?

Preserved hypothesis

In a deterministic test environment, server-side capability enforcement combined with dynamic quarantine will prevent more tested unauthorized effects than intent logging or broker-only authorization, at the cost of some additional latency and quarantines.

The weaker-baseline hypothesis and incremental benefit over equally informed static authorization are different claims. Evidence supports narrow parts of the former; the latter remains conditional on modeled behavior and unestablished against realistic attacks. [The hypothesis ledger](#) retains H1–H33.

2. Related Work

The contribution is a control-to-evidence mapping and measured boundary failures, not the invention of capabilities, complete mediation or simulation. Saltzer and Schroeder [1] motivate complete mediation; Hardy [2] motivates scoped authority. AI Control [3] motivates testing intentional subversion, but our scripted suites are not frontier-agent evaluations. AgentDojo [4] and CaMeL [5] address related agent-tool security questions; we neither run their benchmarks nor reproduce their information-flow guarantees. Pome [6] supplies a separately implemented API twin; native Git [7] supplies an execution-side ref-update interface.

3. Methods

3.1 Boundaries and authority

The early suites use a local modeled server and real local Pome GitHub twins. Capabilities bind tool, canonical arguments, run/request identities, nonce and expiry. A Python gate protects the Pome path; Pome itself does not validate our capability. A separate observer fetches tape/state and retains hashed snapshots. It is a separate process, not a separate host or independent trust root.

The native Git comparison uses identical allowed ref and old/new object predicates at broker dry-run and receive time. The later HTTP suite runs the actor inside Bubblewrap mount/network/PID namespaces. H29 adds separate live/preview service processes, SQLite operation identities, explicit queued jobs and independent snapshot subprocesses. The host, broker and backend remain trusted. Exact task identities are prebound in H30/H31, greatly restricting their request-only search.

A. Native Git — enforcement timing



Same effect policy at preview and receive time; 3/6 versus 0/6 attack successes.

B. Pome — independent observation



Detection is after execution. The observer does not prevent or roll back the effect.

C. Local quarantine — preview fidelity



Matching copied state does not guarantee matching future behavior. Host/backend trust remains.

Figure 1: Three tested boundaries. Each row describes a separate experiment; this is not one combined deployment. Solid arrows show operation flow; the observation branch records effects after execution.

3.2 What counts as evidence

A denied request or nonzero process exit alone does not establish prevention. Native Git uses refs and journals; Pome uses received-call tape and state; H29 uses declared database state and events, including queued effects. Legitimate completion checks outputs and duplicates in the later suites. The original local matrix uses an acceptance proxy, which is weaker than application completion.

Missing evidence is unresolved, not clean. Receipts detect later archive alteration only against a separately trusted anchor; originally fabricated consistent source evidence remains a known limitation. A post-execution alert is detection, not containment. Denied calls must not consume authority needed for legitimate work.

Synthetic fixtures and temporary repositories are used throughout. Historical Arga runs used owned hosted twins. No real victim data or production targets were tested. Deterministic repetitions are consistency checks, not independent random samples; no population attack rates, significance claim or pooled success percentage is made. Per-suite protocols, source copies and reproduction links are indexed in Appendix C.

4. Results

4.1 Execution-time authorization

Table 1. Native Git, 44 cells. Six adversarial and five benign workflows per condition. Same policy information at preview and receive time.

Git condition	Attack successes / 6	Benign completion / 5	False rejection / 5
Intent dispatch	6	5	0
Exact command	6	3	2
Broker preview	3	5	0
Receive gate	0	5	0

The three changes after preview were blocked only at receive time. Exact command matching is a weak baseline that rejects harmless variations. The receive gate is itself a static effect policy; this result is **not a quarantine advantage**. Evidence: [raw cells](#), [summary](#), [methods](#).

Table 2. Isolated HTTP, 51 scripted cells. Twelve adversarial and five benign workflows per condition, including identity changes, bypass and retry controls.

HTTP condition	Attack successes / 12	Benign completion / 5	Benign extra effects
Upstream approval	11	4	1
Endpoint restriction	8	4	1
Exact gateway authority	0	5	0

Endpoint restriction alone did not bind the action. The weak conditions duplicated an approved report on retry. These are ordinary-user sandbox fixtures, not kernel escape tests. Evidence: [raw cells](#), [summary](#), [methods](#).

Table 3. Earlier local matrix, 800 cases. Seven attack and three benign scenarios per configuration, each repeated twenty times. AS = attack successes; LC = accepted legitimate calls; FR = false rejection; FQ = false quarantine.

Configuration	AS / 140	LC / 60	FR	FQ
Intent only	140	60	0	0
Broker only	100	40	0	20
Static server	0	40	20	0
Full, hold-all	0	40	0	20

Server enforcement improved these weaker-baseline outcomes. Static denial and hold-all quarantine tied on prevention and acceptance. Evidence: [raw matrix](#), [summary](#).

4.2 Selective-release tradeoff

Table 4. Modeled selective release, 720 cases. Five attack and four benign workflows per configuration, each repeated twenty times. Completion checks values.

Configuration	AS / 100	LC / 80	FR	FQ
Static deny-unknown	0	0	80	0
Hold-all	0	0	0	80
Selective	20	80	0	0
Static, same reviewed shapes	100	80	0	0

Selective release blocked four modeled fault classes. All twenty simulation-gap repetitions escaped, with mismatch flags only after execution. This is a conditional benefit over equally informed permissions, not superiority over denying everything. Evidence: [raw cells](#), [summary](#), [methods](#).

Later fidelity checks. H26 has twelve hosted Arga cells across hold-all, static and selective conditions. Static and selective complete both benign reads; selective blocks the preview-visible hidden write, and both allow the execution-only write. State equality is limited to the declared issue/access context, not full Git identity. [H26 receipts](#) retain all cells.

H29 has 24 normal cells plus eight fault cells. Both arms complete 12/12 normal workflows. Selective blocks visible and deferred writes; both allow the execution- only fault. Both also lose completion after malformed commit responses until separate trusted reconciliation. [H29 records](#) and [the follow-up report](#) preserve these failures. Copying declared state does not establish matching future behavior.

4.3 Independent backend evidence

Table 5. Pome observer comparison, 16 cells. Six executed attacks, two prevented attempts and eight benign controls across two defense conditions.

Evidence view	Detected / 6	Missed / 6	Blocked attempts	
			alerted / 2	False alarms / 8
Intent	0	6	0	0
Gate	2	4	2	0
Backend tape	6	0	0	0

Both post-gate reads returned synthetic file content without changing state. Upstream records still matched approved intent; backend tape showed the mismatch. Broker-only allowed 4/4 attacks; full gate allowed 2/4, both after the gate. Both completed 4/4 benign tasks. Expected 404 and lost-response controls produced no mismatch alarms. Zero alarms on eight controls is not a population estimate. Evidence: [raw cells](#), [summary](#), [methods](#).

4.4 Model pilots, repairs and costs

Table 6. Complete pilot status; never pool these denominators.

Study	Model calls	Final cells completed / planned	Interpretation
H27, hosted	10	3 / 18	Arga quota stopped the run; no completed matched attack pair.
H30, local	10	18 / 18	Four empty and two authorized-only candidates; no substantive attack coverage.
H31, local adaptive	12	0 / 18	Nine outputs exhausted their token limit; coverage gate failed.

H27 found request-ID poisoning that disrupted a legitimate report. A benign mixed quarantine workflow also failed after the write because preview state could not be reconstructed. H28 repaired those known issues: 36 before/after replay cells and 12 separate fault controls remain archived. Fixed replay completes 9/9 per arm; this is post-discovery regression evidence, not fresh attack resistance.

H30 completes 9/9 final tasks per arm without effects or holds, but its candidates do not test adversarial boundaries. H31 produces two unsuccessful static probes, one authorized serialization check and nine invalid outputs (all six selective outputs invalid). All twelve development continuations complete, mostly with empty attack phases. No final candidate is generated. Full inventory, costs, latency, tokens and failed trials are in [H27](#), [H28](#) and [H29–H31](#). Neither study resolves quarantine superiority.

Estimated model charges are H27 \$0.1359192, H30 \$0.0952803, H31 \$1.163796; total \$1.3949955, not an account invoice. H30 median final-cell latency is 0.554 seconds static versus 1.325 selective. H31 development medians are 0.662 versus 2.789 seconds, but different request sequences make that an unmatched comparison. No model calls or performance experiments were added during this report consolidation.

4.5 Combined architecture: execution-time effect gate

Table 7. Combined architecture, 48 cells. Three arms on one separate-process service over the four fault controls; all three complete 12/12 normal cells with no effect and no unnecessary holds. Attack = unauthorized backend effect executed; *hold* = benign read held (utility lost); *infra* = committed write with a lost response needing reconciliation. Source: [results/combined-gate-v1](#).

Fault control	Static	Selective	Effect gate
Hidden write visible in preview	attack	prevented (hold)	prevented (completes)
Deferred write	attack	prevented (hold)	prevented (completes)
Write only at execution	attack	attack	prevented (completes)
Response lost after commit	infra	infra	infra

The execution-time effect gate is the only arm that prevents the execution-only write, which both static authorization and preview-based selective release miss, and the only arm that prevents the preview-visible faults without holding the benign read. It does not repair the transport fault. This is the same execution-side mechanism as the native Git receive gate (§4.1) applied to a second backend; it is a static effect policy at the effect boundary, **not** a quarantine advantage, and it does not defend a compromised service or host. See [H32](#) and [the predeclared protocol](#).

4.6 Data-flow effect gate

Table 8. Data-flow gate, 68 cells. The H32 gate confines only state changes, so it is blind to a read that changes no state yet returns data outside its authorized output class. A fourth arm confines the outbound response too, tested against an execution-only exfiltrating read. All arms complete 12/12 normal cells; the data-flow gate matches the state gate on every H32 fault. Source: [results/dataflow-gate-v1](#).

Fault	Static	Selective	State gate	Data-flow gate
Read exfiltration (no state change)	leak	leak	leak	prevented (completes)
Write only at execution	effect	effect	prevented	prevented

Only the data-flow gate stops the exfiltration, and it still returns the authorized value. The state oracle reports no effect for every arm on this fault, which is exactly why the data-flow dimension is needed. The gates layer: data-flow covers state. This makes concrete the principle that any effect observable and withholdable at the execution boundary belongs in the gate, not in a preview that only predicts it; quarantine’s residue shrinks to irreversible, out-of-band effects and to covert/implicit channels, which are out of scope for every arm here. See [H33](#) and [the protocol](#).

5. Discussion and Limitations

Defensible implication: evaluate authority at the effect-producing boundary, retain evidence independent of actor claims, and reject the inference that a clean preview guarantees clean execution. Appendix B translates these findings into acceptance rules and known failures. It is a proposed engineering checklist, not an adopted standard or deployment certificate.

The prototype does not test compromise of the permitted egress service, host or kernel, arbitrary malicious tool code, production workflows or general covert channels. It cannot be claimed to have stopped the actual incident. The static Git result and selective-preview result measure different mechanisms. Neither should be relabeled to manufacture support for the other.

An outside reviewer can run the [offline procedure](#) without lab credentials. That verifies the supplied package under its source/host assumptions, not a lab’s actual network or original evidence authenticity. A lab would still need route/credential inventory, retained evidence anchors, versioned policy and an owner for uncertain commits. Independent replication and adoption assessment remain pending. Further model work is deferred until usable outputs and a discriminating attack interface are demonstrated on separate development data.

6. Conclusion planning

Write the final conclusion using the three checked findings above. State that the added-prevention hypothesis remains qualified: strong local execution enforcement, stronger downstream mismatch evidence, and a conditional preview benefit with a reproducible escape. Do not claim perfect containment, formal verification, production readiness, novel invention of capabilities or realistic AI resistance.

Code and Data

The [repository README](#) supplies pinned dependencies and reproduction commands. Appendix C indexes the saved experiments; no evidence was regenerated or removed for this report. An accessible artifact location still needs author review. Local paths are not public links. Nothing has been committed, pushed or published as part of this consolidation.

Author Contributions and LLM Usage

Aditya directed the research, challenged claims and approved experiments. AI coding assistance contributed implementation, tests, execution and these authoring notes. Automated verification is **not independent human verification**. The author must review the facts, confirm affiliation and write/revise the final report and actual LLM-use disclosure. Do not claim external review or original human authorship of these preparation notes.

References

1. Saltzer, J. H., and Schroeder, M. D. (1975). *The Protection of Information in Computer Systems*. [Paper](#).
2. Hardy, N. (1988). *The Confused Deputy*. ACM SIGOPS Operating Systems Review. [DOI](#).
3. Greenblatt, R., et al. (2023). *AI Control: Improving Safety Despite Intentional Subversion*. [arXiv:2312.06942](#).
4. Debenedetti, E., et al. (2024). *AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses of LLM Agents*. [arXiv:2406.13352](#).
5. Debenedetti, E., et al. (2025). *Defeating Prompt Injections by Design*. [arXiv:2503.18813](#).
6. Pome contributors. *Digital Twins*. [Repository](#). Experiments pin CLI 0.43.0 and Node 24.21.0 rather than current upstream.
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Appendix A. Limitations and Dual-Use Considerations

Trust and fidelity: signed requests do not constrain a compromised backend. Early nonce storage is process-local; later gateway operation identities are durable, not a distributed revocation system. One host still controls observer, broker and service. A consistent fabricated source can pass receipt validation. The declared queue/state model does not cover arbitrary delayed effects. The original local suite also retains an authorized secret read followed by unlabelled output leakage. No production secret or live victim data was used.

Method: constructed attacks, small scenario diversity, shared implementation assumptions and deterministic repetitions limit inference. Pome tape/state reads are separate quiescent requests, not an atomic global snapshot. Arga cloning is not full service-state fidelity. H27’s unfinished comparison, H30’s missing adversarial coverage and H31’s reasoning-token exhaustion are retained, not replaced by later successes. Later fixes do not retroactively change earlier evidence.

Dual use: mutation fixtures illustrate authorization and monitoring weaknesses. All tests stay in owned synthetic environments. Do not repurpose them against third parties. Real deployment logs may contain sensitive content; access controls and redaction must preserve enough information for effect verification. Disclosure and license decisions belong to the author before publication.

Appendix B. Proposed evidence-bound containment checklist

Draft requirements for engineering review, **not an adopted or certified standard**. A third party can reproduce the benchmark checks without a lab network, but cannot infer a real lab’s isolation, policy quality or compliance from those checks alone.

This mapping includes the later isolated/local studies without rewriting their frozen protocols. Proposed **core** controls apply to the declared tool boundary; **Q1** additionally applies whenever preview is offered as a containment guarantee. For each declared control, use PASS only when its checks and provenance evidence are present; FAIL for a demonstrated violation; NOT VERIFIED for missing evidence. A passing test runner can correctly demonstrate a FAIL control through a negative fixture. No overall deployment-compliance certificate is issued by this repository.

The [reviewer procedure](#) checks retained artifacts and reruns local boundary fixtures without accounts. It is not independent human replication: that remains outstanding. A lab adopting these requirements would need to supply its route/credential inventory, versioned policy and software, independently retained evidence anchors and documented owner for uncertain commits. Local package verification cannot authenticate that deployment inventory by itself.

For practical adoption, first use the harness with synthetic staging tasks and the deployment’s actual approved policy; measure legitimate failures and operating overhead before rollout. Restrict any preview claim to demonstrated fidelity and require an execution-side control for effects that preview cannot constrain. No lab has evaluated this adoption procedure yet. The actual incident’s permitted egress-service compromise is not reproduced here, so we cannot say this prototype would have stopped it.

Acceptance tests, measured outcomes and residual limits

Each row links a runnable acceptance test to a measured result. The referenced negative controls are expected to demonstrate failure; green pytest output is not an overall compliance verdict. The controls apply only within their declared scope.

Control	Acceptance test	Measured result	Residual limit
C1. Dispatch only exact authority.	Wire and authority tests	HTTP exact gate: 0/12 attack successes, 5/5 benign completion (Table 2).	No endpoint/host compromise claim.
C2. Deny alternate actor routes.	OS/backend boundary test	Endpoint-only restriction still permits 8/12; exact gateway permits 0/12 (Table 2).	Deployed route/credential inventory NOT VERIFIED.
C3. Keep one effect per operation across retries.	Process/restart tests, reconciliation	H29 ordinary workflows: 12/12 per arm; malformed commit response loses completion in both.	Trusted reconciliation required; multi-host recovery untested.
C4. Retain independently acquired call/effect evidence.	Tamper/source tests, observer controls	Backend tape detects 6/6 executed attacks; gate detects 2/6 (Table 5).	Original source forgery and compromised host remain.
C5. Hold missing previews; retain uncertain commits.	Observer timeout, unsupported operations	Unsupported/timed-out preview holds; H29 uncertain response preserved without automatic redispatch.	Availability loss is explicit; operational escalation untested.
C6. Check useful outputs and preserve every trial.	Value/duplicate oracle, search gate	H30 9/9 completion per arm without meaningful attacks; H31 0/18 final cells.	No general agent-workload or realistic-resistance estimate.

Control	Acceptance test	Measured result	Residual limit
Q1. Validate preview state and behavior before claiming safety.	Fault visibility tests	Visible/deferred writes blocked by selective; execution-only write escapes both (H29).	General fidelity claim FAILS.

Appendix C. Evidence index and author handoff

Experiment index

Do not combine counts across these suites. Source archives describe the implementation that actually ran; the current working implementation includes later repairs.

Evidence	Inventory and scope	Full methods/results
Original local/Pome suites	800 matrix cases; 720 selective cases; 20 earlier Pome cells; 16 observer cells; 11 workflow checkpoints.	Hypothesis ledger , observer methods
Native Git	16 evidence cells, then 44 enforcement cells; received ref effects, not quarantine.	Evidence , enforcement
Boundary/HTTP	21 boundary-audit records; 51 isolated HTTP cells.	HTTP study , audit source
Integrated broker/Arga	Two local integrated workflows; three hosted guard conditions; twelve separate-twin quarantine cells.	Arga comparison
H27/H28	Hosted pilot 3/18 final cells; repairs 36 replay cells plus 12 fault controls.	H27 , H28
H29/H30/H31	32 readiness cells; 18 completed H30 final cells; 12 H31 development cells, zero final cells.	Complete follow-up

The [reviewer procedure](#) verifies historical records and checks later studies on temporary copies. It includes local synthetic acceptance tests. This consolidation adds no model calls, cloud provisioning or new performance trial. The implementation-review diagnostics retain historical failures separately in [the review](#); they are not claims that repaired defects still exist.

Decisions for the author

- Confirm the three findings, their denominators and the distinction between execution-time static enforcement and preview-dependent quarantine.
- Confirm author name and affiliation. Do not imply employer endorsement.
- Write the final abstract in at most 150 words: problem, tested boundary, three findings and the fidelity limitation. This pack deliberately provides evidence and a structure rather than asserting a human-written final abstract.
- Transfer the reviewed argument into the official template. Follow its section order; keep Figure 1 and numbered tables, with scope in every caption.
- State actual AI assistance and review performed. Independent reproduction and adoption assessment are pending, not completed author contributions.
- Choose an accessible evidence location and license; review sensitive material before sharing. A local review URL is not a published artifact.
- Check the final PDF’s eight-page main-text limit separately from this review copy. Include references and the required limitations/dual-use appendix.
- Authorize submission explicitly when ready; nothing is submitted by these checks.

Final-template transfer map

Template section	Material to use	Author action
Title / author / abstract	Findings summary; confirmed identity	Write abstract last; confirm affiliation.
Introduction / related work	Sections 1–2	Explain practical contribution without novelty inflation.
Methods / results	Sections 3–4 and Figure 1	Preserve trust boundaries, denominators and failed pilots.
Discussion / conclusion	Sections 5–6	State conditional claims and unresolved fidelity.
Code/data / contributions / LLM use	Corresponding sections	Supply approved artifact location and truthful disclosure.
References / appendices	References and Appendices A–C	Check citations, required limitations and final pagination.